Determine the Whole Given the Part and Percent

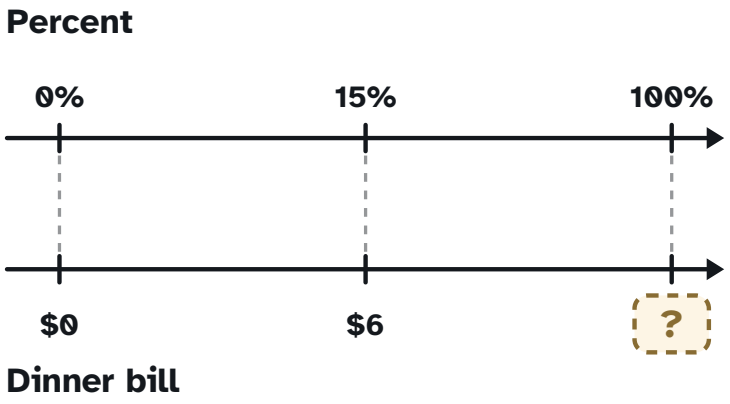
Find the whole when you know a part and its percent, using an equation.

Name _____ **Date** _____

Tonight's spark

A dinner receipt got torn. You can only read the tip: $6. A note says the tip is 15% of the bill. What was the bill?

Make a guess first: did the family spend more than $20, or less?



Our class model: the percent equation

Try it

Work it out on paper. Write one answer on each line.

1 **Which number sentence matches '12 is 40% of a number n'?**

A. 12 × n = 0.4 **B.** n = 12 × 0.4 **C.** 0.4 × n = 12 **D.** n ÷ 0.4 = 0.12

My answer _____

2 **In 0.5 × w = 8, which operation finds w?**

A. Divide 8 by 0.5 **B.** Multiply 8 by 0.5 **C.** Add 8 and 0.5 **D.** Subtract 0.5 from 8

My answer _____

3 **Solve 0.5 × w = 7.**

A. 3.5 **B.** 35 **C.** 7.5 **D.** 14

My answer _____

4 **Solve 0.1 × w = 9.**

A. 0.9 **B.** 90 **C.** 9.1 **D.** 900

My answer _____

5 **Solve 0.25 × w = 6.**

A. 1.5 **B.** 31 **C.** 24

My answer _____

6 **A $9 tip is 20% of the bill. What was the bill?**

A. $45 **B.** $1.80 **C.** $29 **D.** $180

My answer _____

Talk about it

Is the whole bigger or smaller than the part, and how does your check show the answer is right?

Home mission

Use a real receipt. Pretend the tip was $5 and it was 20% of the bill. Can your student write 0.2 × b = 5 and solve it? Then check: is 20% of the answer really $5?

If we get stuck, ask

- Which part of "percent × whole = part" is missing here?
- Should the whole be bigger or smaller than the part?
- Can you check by finding that percent of your answer?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/4-5-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it